

1. $A = 3x^2 + 4x - 4$, $B = x^2 + 3x - 3$, $C = -2x^2 + 3x + 3$ であるとき, 次の計算をせよ。

(1) $B - C$

答.

(2) $2B - 3C$

答.

(3) $A - B + C$

答.

(4) $A + B - C$

答.

(5) $2A - B + 2C$

答.

(6) $3A - 2B + 3C$

答.

(7) $-A - 2B + 3C$

答.

(8) $3A - B - C$

答.

2. 次の 1 次方程式を解け。

(1) $-8x - 1 = -9x + 6$

答.

(2) $6x + 4 = 9x - 9$

答.

(3) $-x - 3(8x - 6) = 5$

答.

(4) $8x - 8 - 3(3x - 6) = 0$

答.

(5) $3x + 2 - 5(7x + 1) = 0$

答.

(6) $\frac{3}{4}x - \frac{7}{3} = -2x + 2$

答.

(7) $-x - \frac{1}{2} = -\frac{1}{2}x - \frac{7}{4}$

答.

(8) $-\frac{5}{4}x + \frac{9}{4} = \frac{5}{3}x + \frac{5}{3}$

答.

1. $A = 3x^2 + 4x - 4$, $B = x^2 + 3x - 3$, $C = -2x^2 + 3x + 3$ であるとき, 次の計算をせよ。

(1) $B - C$

答. $3x^2 - 6$

(2) $2B - 3C$

答. $8x^2 - 3x - 15$

(3) $A - B + C$

答. $4x + 2$

(4) $A + B - C$

答. $6x^2 + 4x - 10$

(5) $2A - B + 2C$

答. $x^2 + 11x + 1$

(6) $3A - 2B + 3C$

答. $x^2 + 15x + 3$

(7) $-A - 2B + 3C$

答. $-11x^2 - x + 19$

(8) $3A - B - C$

答. $10x^2 + 6x - 12$

2. 次の 1 次方程式を解け。

(1) $-8x - 1 = -9x + 6$

答. $x = 7$

(2) $6x + 4 = 9x - 9$
 $-3x = -13$

答. $x = \frac{13}{3}$

(3) $-x - 3(8x - 6) = 5$
 $-x - 24x + 18 = 5$
 $-25x = -13$

答. $x = \frac{13}{25}$

(4) $8x - 8 - 3(3x - 6) = 0$
 $8x - 8 - 9x + 18 = 0$
 $-x = -10$

答. $x = 10$

(5) $3x + 2 - 5(7x + 1) = 0$
 $3x + 2 - 35x - 5 = 0$
 $-32x = 3$

答. $x = -\frac{3}{32}$

(6) $\frac{3}{4}x - \frac{7}{3} = -2x + 2$
両辺に 12 をかけて, $9x - 28 = -24x + 24$

答. $x = \frac{52}{33}$

(7) $-x - \frac{1}{2} = -\frac{1}{2}x - \frac{7}{4}$
両辺に 4 をかけて, $-4x - 2 = -2x - 7$

答. $x = \frac{5}{2}$

(8) $-\frac{5}{4}x + \frac{9}{4} = \frac{5}{3}x + \frac{5}{3}$
両辺に 12 をかけて, $-15x + 27 = 20x + 20$

答. $x = \frac{1}{5}$